

Optimal battery preheating in critical subzero ambient condition using different preheating arrangement and advance pyro linear thermal insulation

Virendra Talele^{a,*}, Uğur Morali^b, Mahesh Suresh Patil^c, Satyam Panchal^d, K. Mathew^e

^a Department of Mechanical Engineering, MIT School of Engineering, MIT ADT University, Pune, Maharashtra, India

^b Department of Chemical Engineering, Eskisehir Osmangazi University, Eskisehir 26040, Turkey

^c Eletromobility, Volvo Group, Gothenburg, Sweden

^d Department of Mechanical and Mechatronics Engineering, University of Waterloo, Ontario, Canada

^e Independent author

ARTICLE INFO

Keywords:

Battery thermal management system
Low temperature
Lithium-ion battery
Preheat
Pyro lining

ABSTRACT

Low temperatures have a substantial impact on the overall performance of traction batteries (0°C and below) as a result, it is essential in developing an effective battery preheating that can efficiently heat up batteries and aid in the start-up of electric cars or energy storage in cold climates. This study explores three different heating methods to find the optimal placement of heating films. Furthermore, based on optimal placement of heating films, we introduced wall-enhanced pyro lining thermal insulation to store potential generated heating temperature inside the battery pack for a longer duration even though power supply from heater is cut-off. The achieved results suggest's, heating model 1 (front face heater) provides uniform heating as compared to heating mode 2 (side face films) and heating mode 3 (bottom face films). In addition to the heating modes, the battery thickness, heat transfer coefficient (HTC) and heating power are explored in the article. The conclusive results show, with heating method 1 time required to achieved 25°C from initiating battery temperature low as -20°C required 624s along with ΔT of 3.51°C, heating method 2 required 395s with ΔT of 33.19°C, and heating method 3 required 290s along with ΔT of 59.04°C. Similarly from the parametric and statistical investigation over the influence of HTC on required heater energy, we conclude that by increasing the amount of HTC, energy required by heater also increases qualitatively as suggested with HTC of 5 W/m²K, 624s of time is required along with 1020 J of energy; for HTC of 10W/m²K, 700s is required with 1150 J of energy; and for HTC of 15 W/m²K, 765s is required with 1280 J of energy. Similarly for the objective of storing maximum generated heat inside the volumetric domain of battery pack when heater power is cut-off at 25°C, the internal temperature of generic model drops by 20°C by the end of 1000s, on other hand improved model drops only 4°C of temperature by the end of 1000s.

1. Introduction

Electric cars have advanced at a tremendous pace during the last several years. Research work has been carried out in many nations to enhance all elements of electric vehicle technology to minimise urban pollution, and promising findings have been obtained as a consequence of this effort. The performance and cost of an electric vehicle are heavily influenced by the performance and service life of the battery it is equipped with. Lead-acid, nickel-cadmium, nickel-metal hydride, lithium-ion, and supercapacitor batteries were the most common battery

systems used in electric cars for primary as well as secondary propulsion [1,2]. As other battery chemistries fade away easily, lithium-ion is gradually taking over as the most common battery technology found in electric vehicles as primary source of propulsion. This is due to the various advantages of lithium-ion batteries, which include high power, high energy density, longer cycle life, low self-discharge rate, longer shelf life, and low pollution [3,4]. The C rating of a battery is the current at which it is charged and discharged. A battery's capacity is often rated and labelled in terms of 1C rate (1C current), which indicates that a fully charged battery with a capacity of 10 Ah (Amp hour) should be capable of providing 10 Amps for one hour [5].

* Corresponding author.

E-mail address: virendratalele1@gmail.com (V. Talele).

<https://doi.org/10.1016/j.tsep.2023.101908>

Received 2 February 2022; Received in revised form 8 March 2023; Accepted 12 May 2023

Available online 13 May 2023

2451-9049/© 2023 Elsevier Ltd. All rights reserved.

Nomenclature			
Symbol	Description	k_x, k_y, k_z	Orthotropic battery thermal conductivity along with x,y,z direction
Ah	Current withdraw capacity [Ah]	Thk _b	Battery thickness
C-rate	Current Charge/discharge [1/h]	Thk _h	Heater Thickness
C _p	Specific Heat [J/kgK]	Thk _p	Pyro block lining thickness
H	Total Enthalpy [J]	W _b	Battery total width
I	Current [A]	W _h	Heater width
k	Thermal conductivity [W/mK]	W _p	Pyro block lining width
L _f	Latent heat of fusion [kJ/kg]	x,y,z	Cartesian coordinate directions
ρ	Density [kg/m ³]	Abbreviations	
Q̇	Heat Generation rate [W/m ³]	BC	Boundary condition
R	Resistance [Ω]	BTMS	Battery Thermal Management systems
T	Temperature [°C or K]	C-rate	Charge/Discharge rate
t	Time [s]	CPCM	Composite phase change material
V	Voltage [V]	DOD	Depth of discharge
W	Power [W]	EV	Electric Vehicle
Subscripts		FVM	Finite volume method
Ah	Current withdraw capacity [Ah]	HEV	Hybrid electric vehicle
chg	Charge	HTC	Heat transfer coefficient
dchg	Discharge	LIB	Lithium-ion battery
H _b	Battery total height	LFP	Lithium iron phosphate
H _p	Pyro block lining height	PCM	Phase change material
H _h	Heater height	SOC	State of charge
H ₀	Heat transfer coefficient	TR	Thermal Runaway
		TMS	Thermal management system

Owing to their high energy density, extended cycle life, and low discharge rate, lithium-ion batteries (LIBs) have become more popular in electric vehicles (EVs) [6]. The performance of the battery, on the other hand, degrades dramatically at low temperatures, particularly below 0°C [7]. The conductivity of the electrolyte degrades, the solid-state lithium diffusion rate degrades, and the charge-transfer resistance between electrolyte and electrode increases for LIBs when operated at low temperatures, resulting in reduced driving range and significant power loss for electric vehicles (EVs) [8,9]. Using low temperature and high charging rate, Tippmann et al. [10] investigated lithium deposition on the anode surface in the form of metal at low temperature and high charging rate and came to the conclusion that charge flux was a key parameter impacting deterioration. What's more, there's a substantial possibility that the developing lithium dendrites would pierce the separator, resulting in internal short circuits, which will cause major safety issues to arise. As a result, battery warm-up/preheating is critical for the functioning of electric vehicles in cold conditions [11,12] Liang et al. suggests that, during thermal management of battery the temperature gradient should be maintained below 5°C regardless of its about preheating or cooling strategy [13].

The battery warm-up/preheating procedure is also included in the battery thermal research. An optimal battery preheating design must serve both objective, heating up a battery in minimum time and maintained temperature inside pack for longer duration. A comprehensive comparative analysis on the accuracy of prediction by two separate and simplified battery models is presented by Chitta et al. [14], Duan et al. [15] which is a numerical analysis on the three-dimensional temperature distribution of a battery liquid cooling system, which showed that the channel width of the cooling plates has a significant impact on the maximum temperature reached by the battery module's cooling plates using four distinct machine learning models to predict the thermal (temperature) and electrical (voltage) behaviours of Li-ion battery cells.

Some recent research in latent heat storage system with Phase change material (PCM) integration have explored its effectiveness in battery heating concept. For instance, Hallaj et al. [16] proposed a PCM-based battery thermal design and discovered that heat generated during discharge was stored as latent heat by the PCM, which was then released

to the module when the battery temperature dropped below the melting point of the PCM (approximately 45°C). In a new passive thermal management technique, Sasmeto et al. [17] suggested using PCM and an insulator to keep the stack temperature above its freezing point, they were able to relieve the stack from the freeze cycles. The thermal mass of the PCM, according to Ling et al. [18], prevents the battery temperature from increasing fast in a single discharge after a lengthy period spent in a cold environment at the ambient temperature. Zhong et al. [19] suggested a preheating technique for an 18650-type battery module based on a paraffin/graphite composite phase change material and resistance wires, which they described as follows: a) Preheating trials demonstrate that the CPCM (Composite phase change material) cooling/resistance wire preheating coupling system can rapidly preheat batteries, although at a somewhat slower pace and with a little higher energy consumption than the pure resistance wire method. b) The connection mechanism may elevate the internal temperature of the batteries by 40°C in 300s. The preheating power may be increased by increasing the resistance wire length, and the equilibrium temperature can be increased by increasing the current. Meanwhile, the rate of preheating is expedited and the amount of preheating energy used is lowered. With regard to prismatic battery cells, Lei et al. [20] used wide-line metal films on the two largest surfaces of prismatic battery cells to preheat the batteries and discovered that under 90W heating power, the battery pack could be heated from 233.15K to restore 80% of the room-temperature discharge capacity in 15 min. There is no doubt PCM coupled heat films are capable in providing both heating and cooling strategy, but PCM are not flame-retardant material hence during charge condition if battery cell gets trigger due either internal or external short circuit, it propagates heat flames within a millisecond in the entire module. Some recent study shows that organic PCMs, such as paraffin and aliphatic acids, have been employed primarily in PCM-based thermal management reported to owing their low cost and latent heat stability advantage. To increase the thermal conductivity of pure organic PCM, several additives such as extended graphite (EG), carbon fibres, and metal particle-based fillers may be used. Nonetheless, the flammability of organic PCMs raises the potential of combustion and explosion, particularly when thermal runaway (TR) is activated [21] During temperature rise of battery there

is rapid change in its electrochemical properties based on irreversible nature of entropy generation and ohmic loss, this phenomenon can be considered in battery governing equation as suggested in literature [22,23].

Despite the fact that PCM-based thermal management research advances towards increasing the thermal conductivity of PCM which is seen as a promising solution in thermal design over the years, it has one significant flaw: It is intrinsically hazardous owing to the high flammability of PCMs, especially those based on paraffin wax. Indeed, once the battery module's auto-acceleratory exothermic process or thermal runaway occurs, the heat storage capacity of these PCMs promotes thermal propagation which increases every millisecond and won't stop until and unless volume entrapped exothermic gases vent out completely [24]. Notably, the transmission of the transient reaction represents a significant challenge for battery safety, as it often exacerbates thermal hazards and there is need for extensive research to be performed on alleviating generated heat from PCM to prevent catastrophic failure of battery in thermal runaway scenario. In view of this one recent research performed by Zhang et al. [25] had proposed hybrid thermal design consist of battery submerged PCM and liquid cold plate, the results show submerged layer PCM alleviates the generated heat from battery by creating buffer layer, to overcome disadvantage of PCM liquid cold plate is coupled in design which constantly removes generated heat from PCM. The proposed system not only has good thermal management performance under the extreme conditions of the power battery system, but it also solves a problem with conventional PCM-based thermal design that don't work well because of the saturation of PCM heat absorption under the cycle conditions. In addition, the proposed system can effectively slow down or stop the spread of thermal runaway. In passive thermal design an recent emerging heat pipe assisted PCM cooling strategy notes positive advantage in delaying maximum temperature of battery and improving its thermal performance as suggested by Lian et al. and Wang et al. [26,27], where the influence of external ambient temperature over battery cooling performance and temperature delay strategy is extensively studied, the main finding from this literature denotes the battery health is good when its temperature gradient is maintained below 5°C, the novel findings based on the influence of ambient temperature over battery thermal performance is also noted in this studies. A recent study by [28] shows the capability of battery preheating by using liquid immersion strategy for high energy dense prismatic cell, where they have targeted empirical objective of battery preheating by thermal design till its maximum temperature reaches 25°C since after that when vehicle commute on road, battery self-heat itself by change in internal electrochemical energy, however this self-heating of battery is very sensitive and depends on the drive cycle of vehicle hence there is always a need of dynamic battery preheating system which can make sure to preheat batteries and maintain its temperature in efficient range of 25°C to 45°C.

Based on the authors knowledge there is very scant research noted owing to the fact that there is need of concrete battery preheat design which is capable to preheat large format prismatic battery uniformly and able to store generated heat inside thermal design for longer duration even though when the power with preheating device is cut off [29]. No doubt in past literature studies have attempted to solve this problem but those are targeted with high cost preheating thermal design (Liquid immersion fluid flow, Cold plate) and less focused were given for low-cost preheating system for 2-Wheeler and 3-Wheeler battery manufactures where economically stable preheating is indeed requirements. In present study we have proposed low-cost heater assisted with advance pyro lining insulation which is not only able to preheat batteries quickly but also able to store generated heat inside thermal design when power is cut off. The present study is aimed to answer some fundamental questions like: a) During battery preheating which is an optimum arrangement of heater? b) How temperature gradient inside battery is changed during different preheating configuration by heater? c) How fast these high energy dense and large format prismatic battery can be

preheated based upon different preheating configuration? d) How the power and energy consumption by heater plays an important role in different preheating configuration, and most importantly, d) If a BEV is parked in sub-zero ambient temperature and there is no source to warm the battery after commuting off the road, how long can the already created preheat energy and temperature persist in thermal design compared to both without and with thermal insulation?

2. Modeling

2.1. Problem statement

In this research work, we presented a heating film linked pyro lining concept for large format prismatic battery which can be potentially extended on pack level configuration. Our battery pre-heating thermal design was investigated with taking into account on operating strategy in cold ambient conditions, structural compact design, and storing maximum heat within volume by avoiding thermal runaway mitigation by PCM. This approach is capable of maintaining operating temperature homogeneity during heating by enabling stored heat to stay inside the enclosed volume for a longer period of time when the heater supply is interrupted. Because the active materials in lithium-ion batteries are porous media structures, the conductivity of prismatic cells are specified orthotropic in nature [30], as seen in Fig. 1, three different heating concepts are investigated in this research with the goal of determining which one provides better temperature homogeneity and is capable of rapidly heating the cell. Extending this concept of optimal preheating, within the objective of storing generated heat for longer duration of time evnthough power supply from heater is cut-off, we introduce low thermal conductive pyro lining insulation at the wall of battery as shown in Fig. 2 and Fig. 3.

In the present study, three heating modes are explored with the objective to find an optimal heating method that can distribute uniform temperature within a volume of the entire cell domain. Fig. 1 represents three types of heating modes a) Front face sandwich heating model, b) Side face plate heating model and c) Bottom face heating model. Based on the achieved result, we improved heating model by introducing ceramic-based pyro lining [31] at the wall of the heat plate. The predicted advantage of using pyro liners at a wall is to store generated heat inside the volumetric cell domain even after supply to the heating plate is cut off. This method helps driver in cold region countries to store heat inside battery pack and can be utilized when they are not accessible to battery heating provision [39].

2.2. Physics description

The present numerical scheme is developed using commercial coded Ansys software within Ansys fluent 2021 R2 framework. The battery used for the present investigation is 150 Ah LiFePO4 Prismatic Cell with 3.2 V as nominal voltage. Fig. 3 represents geometrical description of battery model. The thermophysical property used in the numerical model is shown in the Table 1.

As discussed in the previous section, the effect of battery thickness, heat transfer coefficient, and pyro lining wall thickness is also going to be investigated in terms of battery heating performance for which the required thermophysical property is shown Table 2.

In battery heating purposes flexible type of polyimide preheater is extensively used due to its flexible bending with high tensile strength mechanical capabilities. This heater can be easily attached to the varying uneven surface of batteries [33]. For insulation purpose we had used pyro block lining material, this material comes under ceramic refractory grades, and it has lower thermal conductivity hence it can be utilized as a perfect insulator at the wall of battery pack enclosure to entrapped generated internal heat inside the battery pack even though power to the heating source is cut off, which typically observed in cruise mode [34]. These pyro liner poses lower thermal conductivity rate additionally they

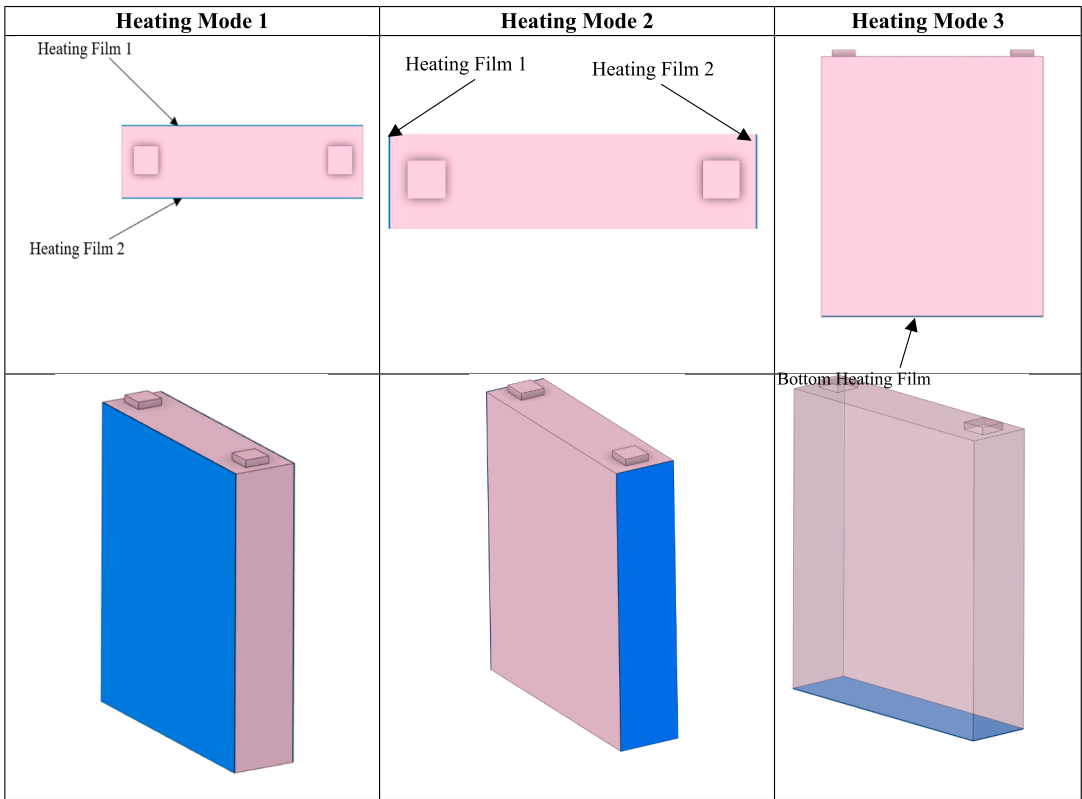


Fig. 1. The Schematic of heating model without pyro lining a) Front face heating lining, b) Side face heating lining and c) Bottom face lining.

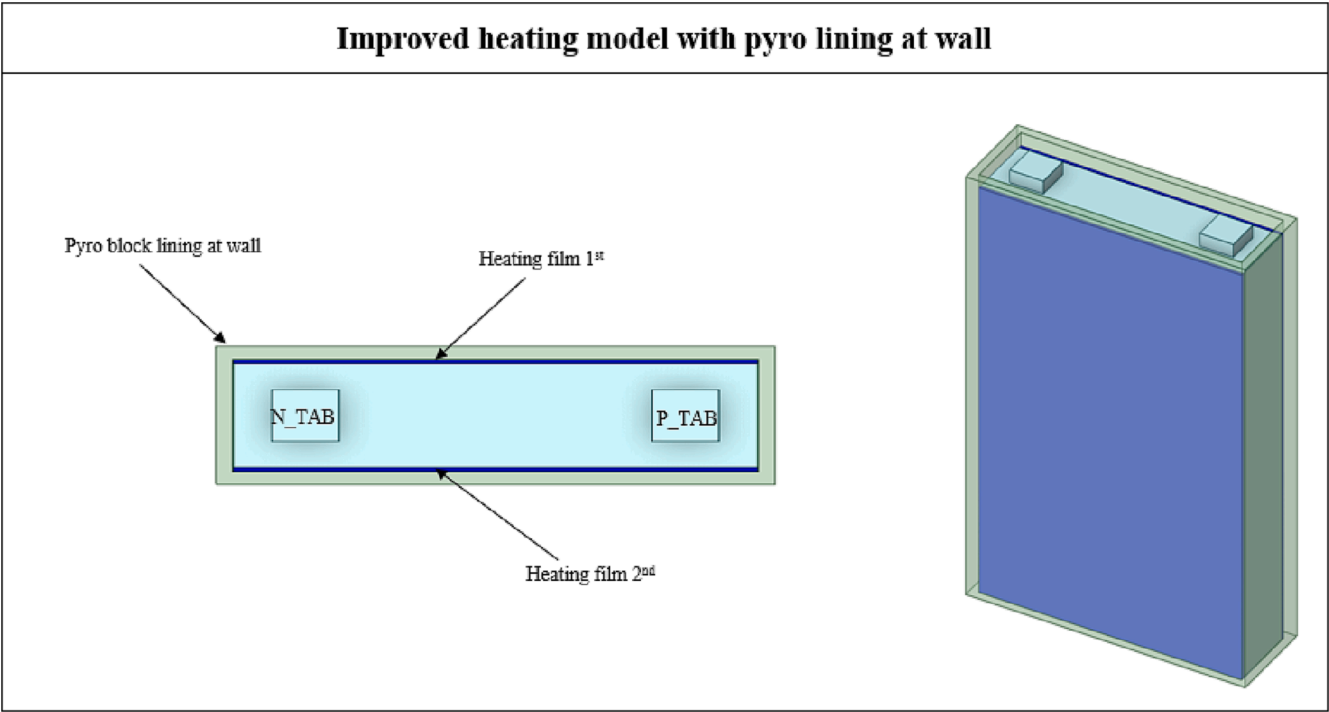


Fig. 2. Improved heating model.

pose higher melting temperature up to 1760°C hence it also suggests potential application in delaying battery thermal runaway propagation. Another advantage of using pyro liners is that they are not prone to catch fire, even ceramic grade material doesn't produce smoke in a fire as per development requirements of Section 803 of the International Building

Code (IBC) for interior wall and ceiling materials. The granular structure of pyro liners is made with a strong covalent bond hence during the case of fire it doesn't melt prematurely and it can provide greater structural stability even in the case of fire events, the noted melting point of the pyro liner is 1760°C which is again 70% greater than onset thermal

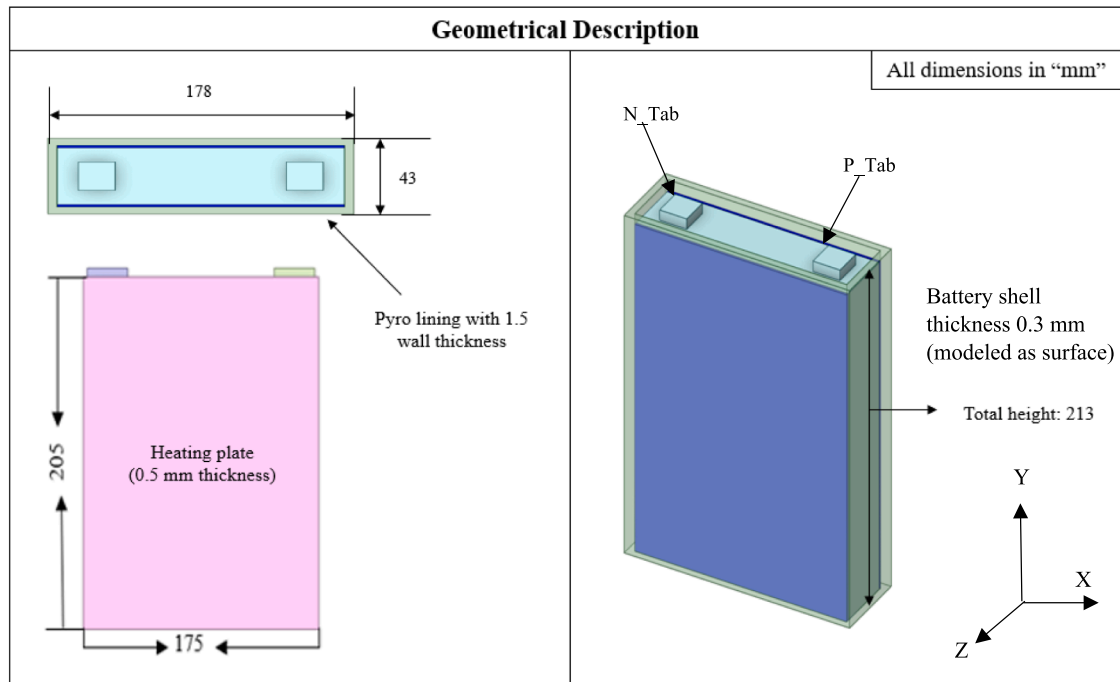


Fig. 3. Geometrical description.

Table 1
Thermophysical property.

Sr No	Parameters	Symbol	Unit	Battery Jelly roll [32]	Battery shell [41]	Positive Tab [41]	Negative Tab [41]	PI Heater [33]	Pyro block lining [34,44]
1	Thermal conductivity	k	W/mK	k _x = 18.4 k _y = 18.4 k _z = 1.2	237	1.04	1.04	15	0.1
2	Specific heat capacity	c _p	J/kg·K	1060	897	800	641	1090	113
3	Density	ρ	kg/m ³	3575	2700	1500	223	1400	240
4	Battery voltage	Volt	V	3.2	–	–	–	–	–
5	Battery Capacity	–	Ah	150	–	–	–	–	–
6	Type of battery	LFP	LipFeO4	Lithium Phosphate	–	–	–	–	–
						cell chemistry reaction [3]			
	Electrode					Electrochemical reaction			
8	Anode					Li _n C ₆ ≈ Li ₀ C ₆ + nLi ⁺ + ne [–]			
9	Cathode					Li _m – nFePO ₄ + nLi ⁺ + ne [–] ≈ Li _m FePO ₄			
10	Overall					Li _n C ₆ + Li _m – nFePO ₄ ≈ Li ₀ C ₆ + Li _m FePO ₄			

Table 2
Property considered influencing simulation.

Sr no	Component	Symbol	Units	Value
1	Battery total height	H_b	mm	210 (Including tabs)
2	Battery total width	W_b	mm	175
3	Battery thickness	Thk_b	mm	39
4	Heater width	W_h	mm	175
5	Heater height	H_h	mm	205
6	Heater Thickness	THk_h	mm	0.5
7	Pyro block lining height	H_p	mm	213
8	Pyro block lining thickness	Thk_p	mm	1.5
9	Pyro block lining width	W_p	mm	178
10	Ambient temperature	—	°C	−20
11	Natural convective heat transfer coefficient	H_0	W/m ² K	5, 10 and 15
12	Heating Power	—	W	20

runaway temperature of battery, overall pyro block thermal insulation provides multiple advantages and can be used for multipurpose application of preheating (Since due to lower thermal conductivity, the sub-zero ambient temperature provides less influence on heating performance) and delaying thermal runaway propagation between cells at pack level configuration, which is also one of the prominent requirements in EV Bus and Truck applications as stated in GB38032 [31,42] (see Fig. 4).

Table 2 indicates the parameters used for the present study with a single objective optimization method. The primary objective of the study was to determine the optimal heating method out of three as suggested in Fig. 1. Once the optimal heating configuration is identified, the next objective is to determine the influence of battery thickness, HTC and pyro block thickness on battery heating application. The initiating ambient temperature is kept constant as −20°C along with the heater power of 20 W.

2.3. Numerical modelling

The conservation of energy in the present transient numerical study is given by below Eq. (1).

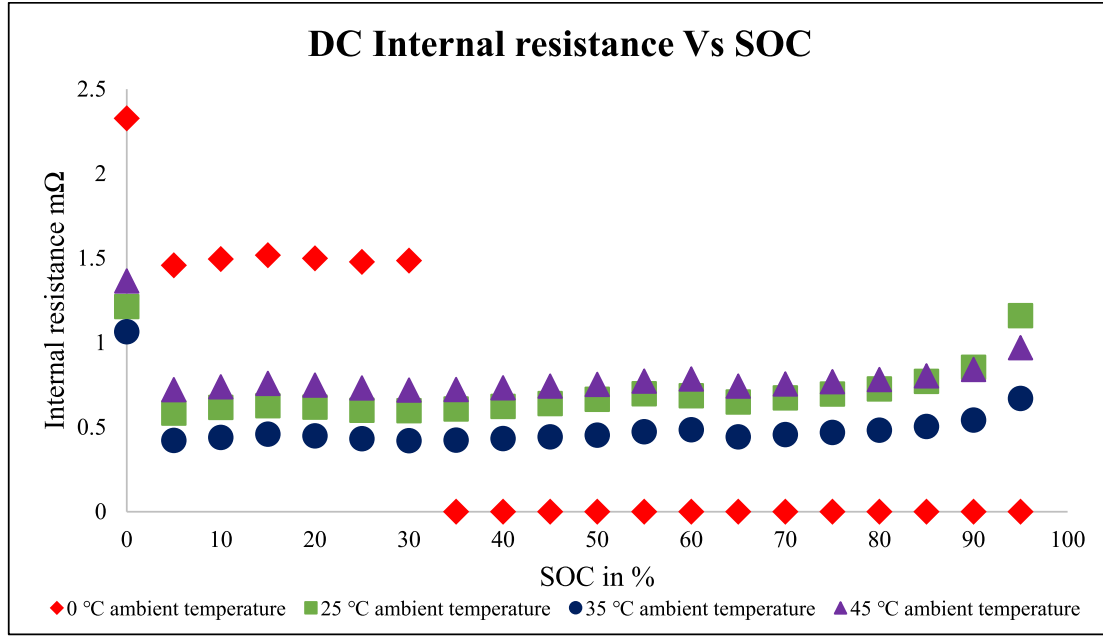


Fig. 4. Internal resistance of battery for different ambient temperature.

$$\rho C \frac{\partial T}{\partial t} = k \nabla(\nabla T) \quad (1)$$

In above Eq. (1); ρ shows the density of material, C is specific heat and T is thermal conductivity. In present study the heating models of battery are assumed to be operated in -20°C ambient condition with submerged in natural convective air domain which is set at boundary. The heating film is very thin about 0.5 mm which are treated as surface and its equation can be expressed below.

$$-k \nabla T|_T = \dot{q}'' \quad (2)$$

In above equation heat flux is expressed by $\dot{q}''$ for heating film. Where $\dot{q}''$ is define with respect to heating power and surface area $\dot{q}'' = P/A$. In present study natural convective boundary condition is set over outer domain and solver is initiated at -20°C ambient temperature which is expressed by below Eq. (3) [35].

$$-k \nabla T|_T = h_0(T_s - T_{amb})|_T \quad (3)$$

where h_0 is the natural convection heat transfer coefficient, T_s is the surface temperature of the battery cell and T_{amb} is the ambient temperature. The natural convective coefficient in present study is taken as $5\text{ W/m}^2\text{K}$, $10\text{ W/m}^2\text{K}$ and $15\text{ W/m}^2\text{K}$ at free stream velocity temperature, because it represents best value of natural convective coefficient as suggested by Kurhade et al. [43].

When battery temperature is increased by preheating method battery also provide its resistance to heating power which can be expressed in terms of battery irreversible ohmic loss equation as expressed below [36].

$$Q_{ohmic} = I(E - E_{oc}) \quad (4)$$

The above Eq. (4) shows Joule or Ohmic heat produce by battery. Here, for making simulation more comprehensive and co-helping we had curve fit ohmic losses data of battery in terms of UDF. The ohmic loss data is acquired from sub-zero to 45°C . Below figure shows the ohmic loss data for present prismatic 150 Ah cell.

The above figure shows, value of internal resistance is highest during sub-zero condition at lowest SOC.

2.4. Mesh independent study

The present research takes use of a programmed tool that is commercially accessible. ANSYS fluent 2021 R2 and Ansys fluent Mesher [36] are used to generate the finite volume mesh. Because the output results are largely reliant on the mesh type and size used, it is required to undertake mesh and grid-independent studies to evaluate the output results' quality. Another critical part of the mesh independent analysis is determining if the produced mesh is consistent with the solution's convergence time. To begin with the numerical scheme, we choose the optimal mesh type that takes the least amount of time to compute without sacrificing the accuracy of the output; hence, three kinds of grids are investigated: a) tetrahedral, b) polyhedral and c) hexacore for the case of heating mode 1 by monitoring average time required to reach 25°C at HTC of $5\text{ W/m}^2\text{K}$ at the boundary. Fig. 5 shows the 3 types of mesh.

As Table 3 results indicate, the polyhedral grid type takes the lowest computational time for the convergence of solution without compromising with the quality of output results. Skewness correlation suggests for polyhedral mesh type it is lowest with a value of 0.72 and aspect ratio of 0.18 which confirms the correct mesh type requirement for Ansys fluent solver. The next goal is to determine correct mesh size for **polyhedral mesh type**. As presented in the Table 4, mesh size of 2 mm is selected and convergence time is calculated with 4 cores 2.6 GHz operational processor.

The mesh independent study indicate, 2 mm is an optimal mesh size selection for the polyhedral grid as highlighted in above Table 4.

2.5. Orthotropic thermal conductivity of cell

As suggested by Zhou et al. [37] and Mussa et al. [38], the thermal conductive property of a prismatic cell composed of numerous potentially charge active jelly roll material linked in sequence inside the outer metallic shell. When energy is transferred from these porous jelly rolls, they behave like an active orthotropic medium with a high thermal conductivity within the in-plane direction X-axis and Y-axis. The value of thermal conductivity is measured minimum along the out plane direction Z-axis because of the resistance offered by jelly roll thickness. Fig. 6 shows the thermal conductivity of a prismatic cell is assigned in a Cartesian coordinate system.

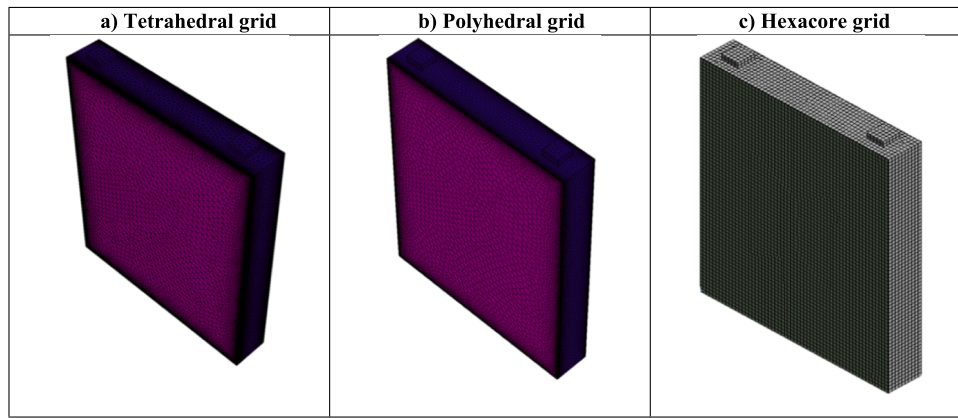


Fig. 5. Types of mesh.

Table 3

Grid selection study.

Sr No	Mesh Type	Skewness	Time required to reach 25°C	Cell counts	Time for convergence (s)
1	Tetrahedral	0.92	626	126,695	528
2	Polyhedral (Selected)	0.72	624	098,564	312
3	Hexacore	0.85	624	105,867	501

As Fig. 6 indicates, the thermal conductivity for the orthotropic cell is maximum for in-plane direction (X and Y), and lowest for out-of-plane direction (Z) indicating the resistance to the heat transfer along the Z-axis.

2.6. Validation of present numerical scheme

The results from present study are validated with Zhang et al. [40], where they have modelled battery preheating system with film based preheater including configuration 1) which have heating film at both side of battery and configuration 2) which have heating film at front face of battery. There results also indicate that with configuration 1 of having battery heating film at dual side face of battery indicates instability in temperature gradient of battery however with the configuration of battery heating film at front face of battery have uniform temperature gradient as shown in below Fig. 7.

Above Fig. 7 of Validation plot shows the accuracy of present model. A similar pattern of non-uniform heating and highest temperature difference is observed in configuration of battery with dual side face mounted heater film, and uniform low temperature gradient is observed in configuration of battery with front face heating method. The concise value suggests for configuration 1 of dual side heating film around battery, the temperature difference was 31°C by Zhang et al. and with present case it is 33.19°C, Similarly for the configuration of front face heating method it was 4°C in Zhang et al. work and in present model it is 3.51°C. The overall error percentage shows, for configuration 1 and 2 between Zhang et al. and present work is less than 7%.

Table 4

Mesh independent study.

Sr No	Mesh size (mm)	Polyhedral Skewness	Time required to reach 25°C	Time for convergence
1	12	1	Divergence	-
2	8	0.99	200 s	98 s
3	4	0.90	600 s	250 s
4 (Selected)	2	0.72	624 s	312 s
5	1	0.71	624 s	550 s
6	0.5	0.701	624 s	910 s

3. Result and discussion

3.1. Comparison between heating methods 1, 2 and 3

The objective of the present section is to select an optimal heating configuration that can maintain heating uniformity within the cell domain and can raise temperature up to 25°C in less duration of time by maintaining optimal thermal gradient within cell domain. The constant heating power of 20 W is supplied from heater, Fig. 8 indicates results from the present section.

The present section deals with the results for different heating methods as explained in Fig. 1. The achieved results indicate in heating method 1, there is an optimal temperature distribution around the

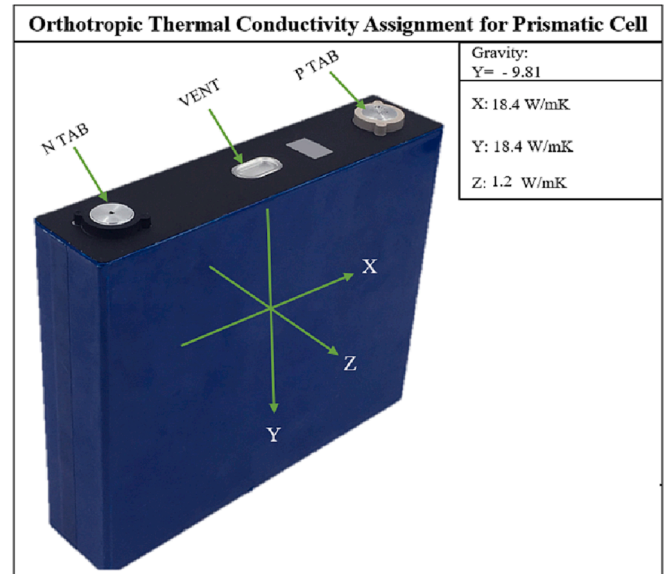


Fig. 6. Assignment of orthotropic thermal conductivity.

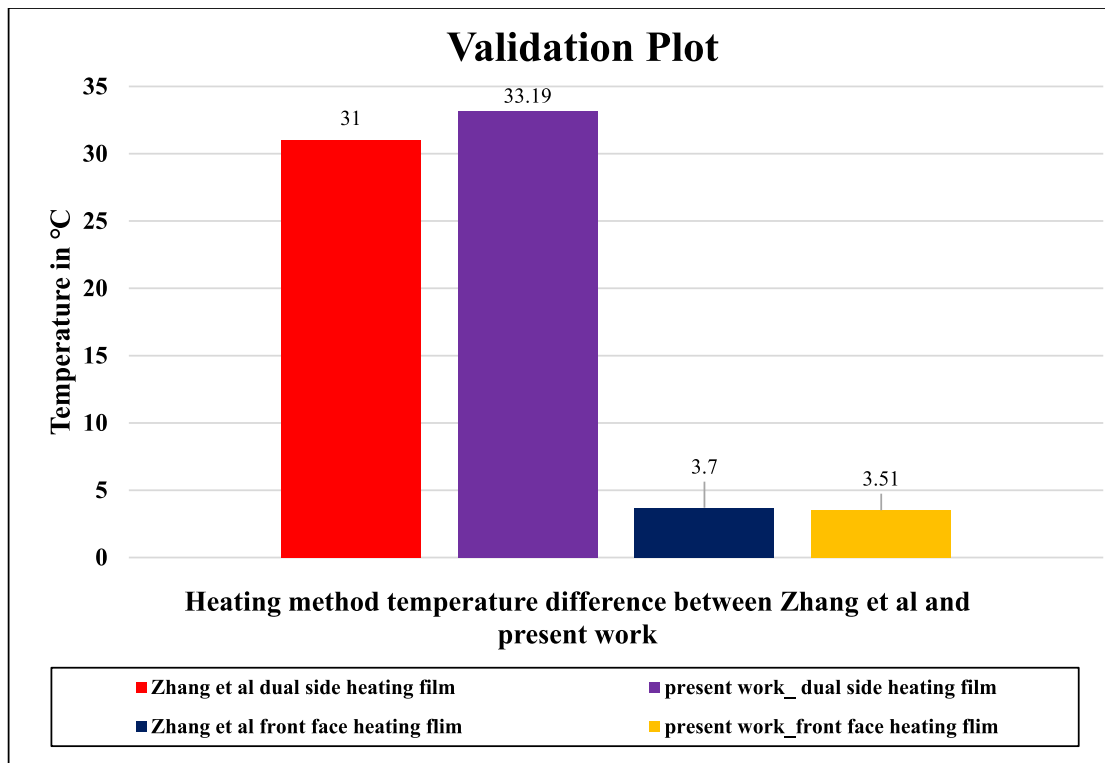


Fig. 7. Validation plot.

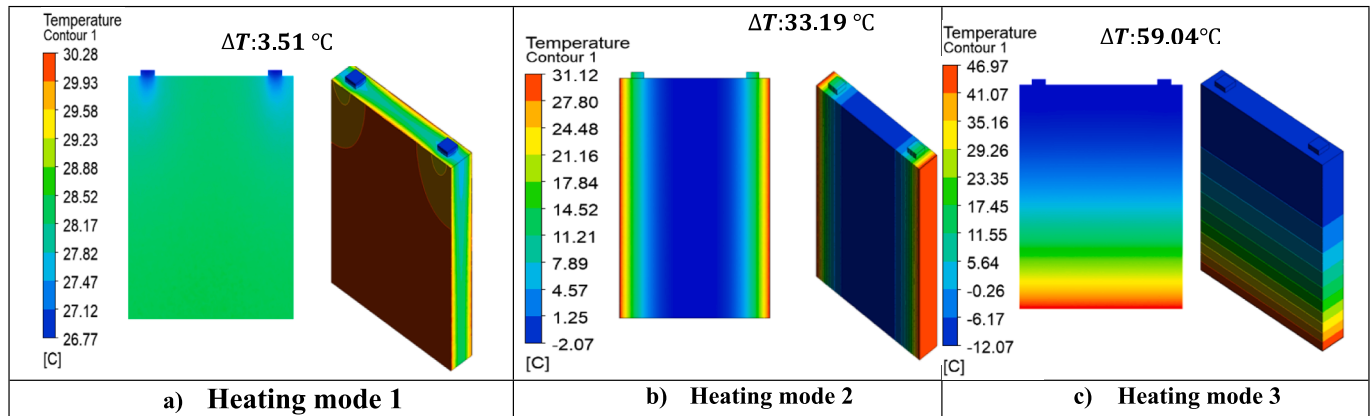


Fig. 8. Heating mode with different configurations maximum volumetric temperature a) Front face heating lining, b) Side face heating lining and c) Bottom face lining.

volume of prismatic cell domain as compared with heating method 2 and 3. The temperature difference ΔT for heating method 1 is 3.51°C , heating method 2 is 33.19°C and heating method 3 is 59.04°C . However, in the time impedance study heating method 1 requires more time to reach 25°C battery temperature as compared to the other 2 methods. Even though heating method 1 required more time to reach threshold limit, it is selected as optimal arrangement owing to its ability to distribute uniform temperature in volumetric domain as compared to other arrangements. Generally, a large tail is seen at the tail of above plots because a lumped modelling strategy is used in present study in

which the temperature of such lumped thermal domains are only a function of time $T = T(t)$. It can be noted that the rise of battery temperature is less against time and once isothermal equilibrium reached in thermal domain the simulation becomes converged. Notably, for the optimal heating method 1, a power of 20 W was chosen because raising the heater power anything above 25 W resulted in a temperature gradient change of over 5°C , which can be detrimental to the health of the battery. As indicated previously, 20 W results in a battery temperature gradient of 3.51°C , while 25 W gives a gradient of 4.89°C . However, during the statistical evaluation of the system, we fixed the upper

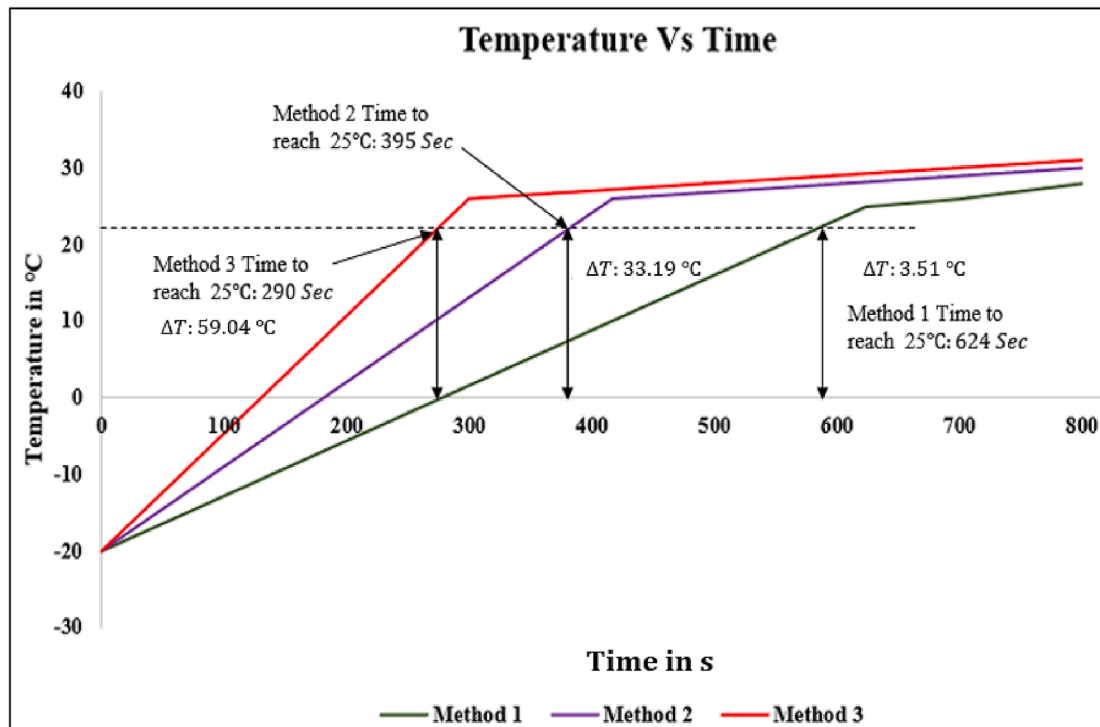


Fig. 9. Temperature Vs Time for different heating methods.

limit of the temperature gradient to 25 W because the parametric analysis proved that with a power of 25 W, the battery temperature gradient remained less than 5°C (see Fig. 9).

3.2. Effect of HTC on heating application

In this section, we had investigated the influence of initiating HTC boundary condition on heating method 1. The purpose of this study is to

determine on how variation in HTC affect heating energy supply and time required by battery to reach 25°C threshold limit.

The achieved result indicates at HTC $5\text{ W/m}^2\text{K}$ 624s time required along with 1020 J heating power consumption, furthermore when HTC is increased time required to reach 25°C threshold also increases along with heater energy requirement, as in for HTC of $10\text{ W/m}^2\text{K}$ time required is 700s with 1150 J of energy consumption and HTC of $15\text{ W/m}^2\text{K}$ required 765s with 1280 J of energy consumption (see Fig. 10).

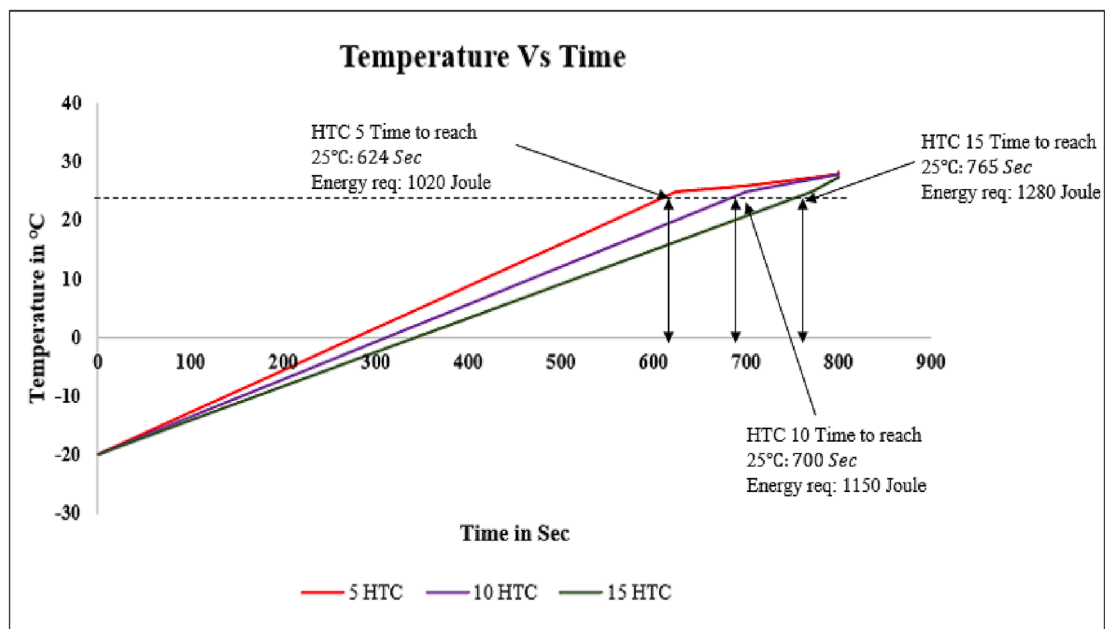


Fig. 10. Temperature Vs time (Impedance of HTC on heating power and time required to reach threshold limit).

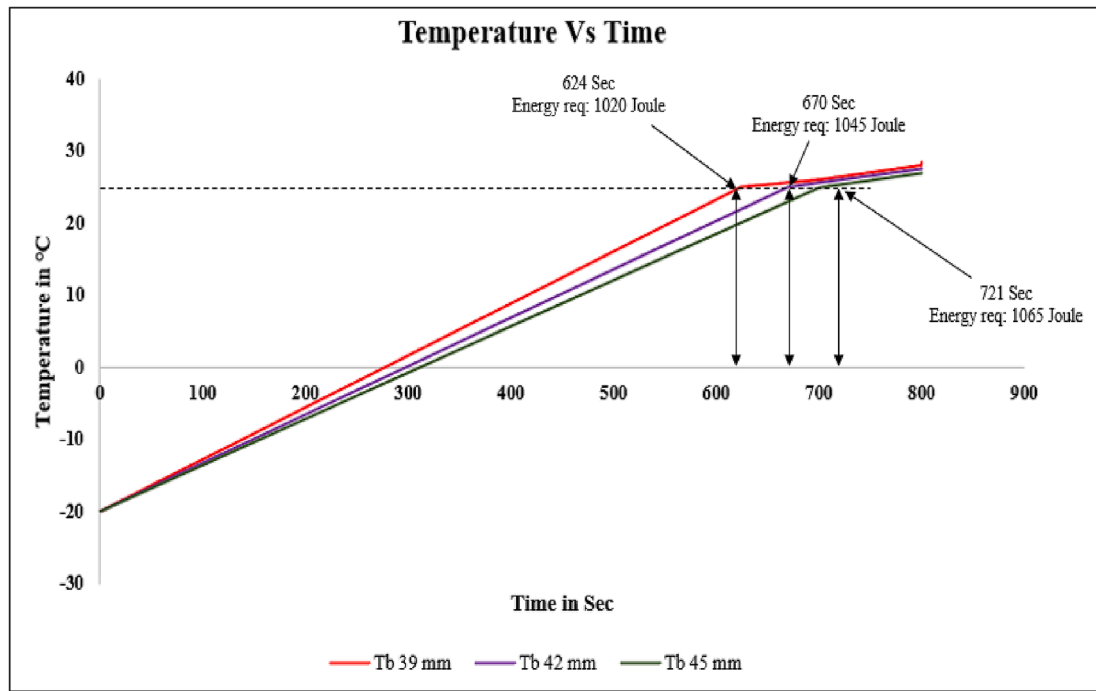


Fig. 11. Temperature Vs Time (Effect of battery thickness of heating power and time required to reach threshold limit).

3.3. Effect of battery thickness on heating power consumption and time required to reach threshold limit

In this section we have investigated the effect of battery thickness on heater power consumption and time required to reach set threshold limit of 25°C. The battery thickness is varied in three cases, case a with 39mm, case b with 42mm and case c with 45mm.

As the results from Fig. 11 indicate when battery thickness is increased the time required for the battery to reach the set threshold limit of 25°C also increases additional with the energy consumption by the heater to reach that temperature. In the case of Thk_b with 39mm, the

time required by the battery to reach the threshold limit is 624s with 1020 J of energy consumption by the heater. For Thk_b 42mm time required is 670s with 1045 J energy consumption by the heater and for Thk_b 45mm, the time required is 721s with 1065 J of energy consumption.

3.4. Improved model effect on heating application

In the previous section, we noted generic heating method 1 provides optimal temperature uniformity as compared to heating mode 2 and 3. However, with heating method 1, the time required for the battery to

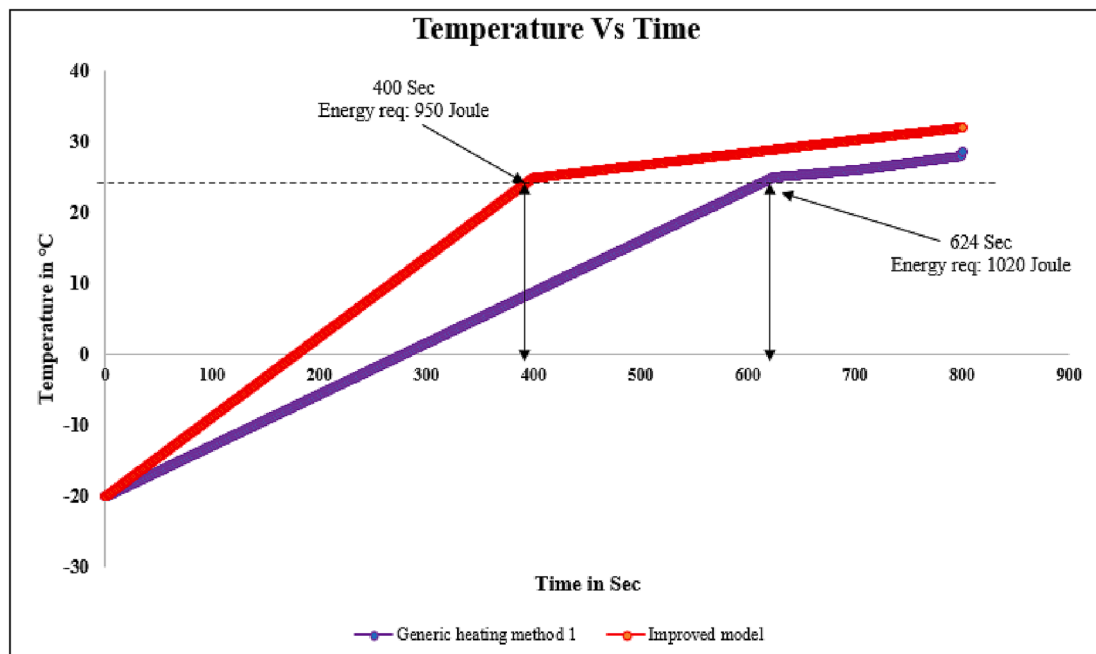


Fig. 12. Temperature Vs Time (Comparative study generic model Vs Improved model on heating power and time required to reach threshold limit).

reach the threshold limit is higher with additional more heating power. To improve this drawback, we introduce ceramic-based pyro liners at the wall by insulating battery inside that control volume. These pyro liners are capable to store generated heat inside geometrical volumetric space as shown in the Fig. 2. This improved model is able heat up batteries with lowest energy consumption as shown in the Fig. 12.

As the results from Fig. 12 indicate, an improved model battery can heat up quickly as compared to a generic heating mode. This improves model also maintained temperature distribution along the interior battery domain with a temperature difference of 3.51°C . The exact time required with the improved model is 400s to reach the temperature threshold limit with 950 J of energy consumption by the heater, comparatively with generic heating method time required to reach threshold limit is 624s with heater energy consumption of 1020 J which is more than the improved model.

3.5. Comparative internal energy analysis between generic heating method 1 and improved wall pyro lining model

In the previous section, we found out that an improved heating model can heat up the battery quickly with the lowest energy consumption by the heater. The next objective is to investigate whether the improved model is capable to store generated volumetric heat inside domain when power supply is cut off. In this section, a detailed comparative analysis for energy storage capacity is discussed between generic heating method and improved model post the cut-off of power supply from the heater.

When the power source of the heater is cut-off, the improved model with thermal insulation is capable to store generated heat inside volumetric space for a longer duration as compared to the generic heating method 1. As the results from Fig. 13 indicate, the improved model reaches the threshold limit of 25°C within 400s and further when the heater power supply is cut-off, the minimum temperature drop is only 4°C . Comparatively, for the case of heating method 1, it requires 624s to reach the threshold limit of temperature and when the power supply is cut-off, there is significant temperature drop of 20°C . Hence improved model is capable to store generated temperature for a longer duration as compared to a generic model. This design justifies our objective of optimal design which can heat up the battery more quickly, by maintaining temperature gradient, consuming lower energy supply from the heater, and being able to store generated temperature for a longer

Table 5

Levels of the factors.

Level	Heat transfer coefficient	Battery thickness	Pyro block thickness	Heating power
Level 1	5	39	1.25	10
Level 2	10	42	1.50	15
Level 3	15	45	1.75	20
Level 4	20	48	2.00	25

Table 6

L16 orthogonal array and response variable.

Run	heat transfer coefficient	battery thickness	pyro block thickness	heating power	time
1	5	39	1.25	10	691
2	5	42	1.50	15	679
3	5	45	1.75	20	632
4	5	48	2.00	25	621
5	10	39	1.50	20	700
6	10	42	1.25	25	708
7	10	45	2.00	10	731
8	10	48	1.75	15	719
9	15	39	1.75	25	719
10	15	42	2.00	20	751
11	15	45	1.25	15	796
12	15	48	1.50	10	832
13	20	39	2.00	15	847
14	20	42	1.75	10	858
15	20	45	1.50	25	876
16	20	48	1.25	20	902

duration.

4. Statistical evaluation

It is significant to evaluate the effect of each factor on the time to reach 25°C . With this aim, we proposed to statistically investigate the effect of a number of factors on the time. The factors were the heat transfer coefficient, battery thickness, pyro block (Insulation) thickness, and heating power. The response variable was the time to reach 25°C .

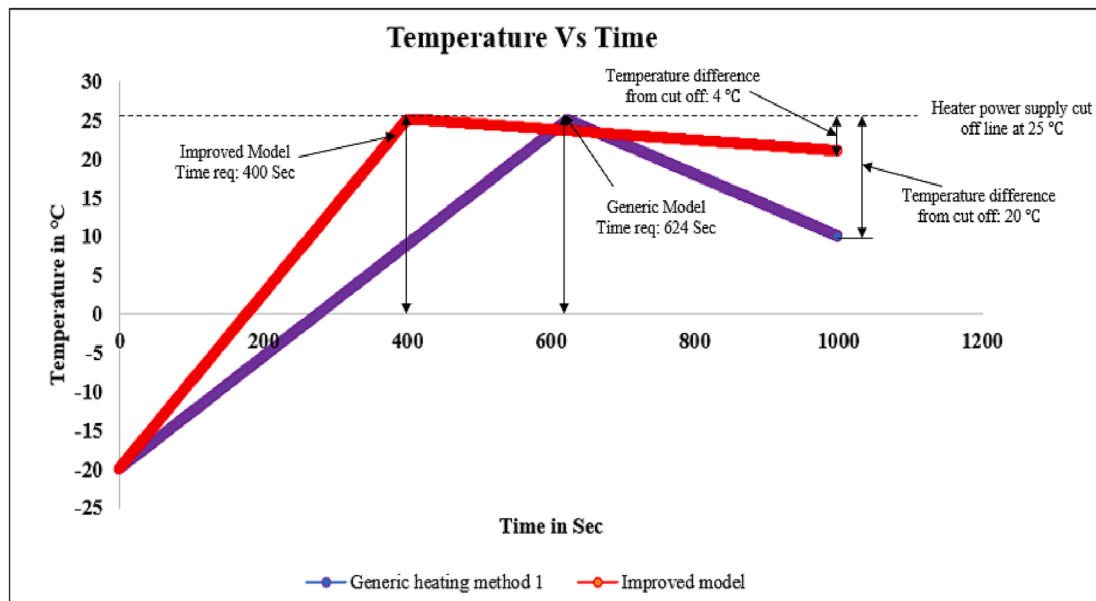


Fig. 13. Temperature Vs Time (Comparative energy storage capacity between Improved model Vs Generic model).

Table 7

Response table (smaller-is-better).

Level	heat transfer coefficient	battery thickness	pyro block thickness	heating power
1	−56.33	−57.35	−57.73	−57.79
2	−57.08	−57.46	−57.70	−57.59
3	−57.77	−57.54	−57.24	−57.38
4	−58.80	−57.63	−57.30	−57.21
Delta	2.47	0.28	0.49	0.57
Rank	1	4	3	2

The levels of each factor are presented in Table 5.

An orthogonal array L16 was created in Minitab by using the levels of each factor. The orthogonal array provided a statistical examination of the effect of each factor separately [45]. The orthogonal array matrix is presented with the response variable in Table 6.

The signal-to-noise (S/N) ratio was calculated using the response variable according to the smaller-is-better quality characteristic. The obtained S/N ratios were used to calculate the S/N ratio of each level of each factor. The S/N ratio of each level of each factor was used to calculate delta values and corresponding rank values [46]. Table 7 shows the response table. For the heat transfer coefficient, the highest S/N ratio and the smallest S/N ratio were −56.33 and −58.80, respectively. The delta value for the heat transfer coefficient was obtained by the difference between these S/N ratios. In other words, the delta value for the heat transfer coefficient was 2.47. The highest delta value obtained for the heat transfer coefficient showed that the effect of the heat transfer coefficient was the highest compared to other factors. Furthermore, the second-highest delta value of 0.57 demonstrated that the effect of the heating power on the time required to reach 25°C was smaller than the heat transfer coefficient and higher than other factors. Table 7 also indicated that the effect of the heat transfer coefficient was approximately four times bigger than that of the heating power. The delta value of pyro block thickness was 0.49. Moreover, the response table showed that the effect of the pyro block thickness was higher than that of the battery thickness and smaller than that of the heating power. On the other hand, the lowest delta value of battery thickness showed the lowest effect of this factor. The lowest delta value of battery thickness showed that the effect of the heating power was two times higher than that of the battery thickness.

Variance analysis (ANOVA) was performed according to the significance level of 95% to determine if there was a statistically significant relationship between the factors and the response variable. Table 8 showed that the P-values of heat transfer coefficient, and pyro block thickness were smaller than 0.05. The P-value smaller than 0.05 demonstrated that there was a statistically significant relationship between these factors and the response variable. Furthermore, the F-value of the heat transfer coefficient was higher than that of the pyro block thickness. The higher F-value of the heat transfer coefficient indicated that the effect of the heat transfer coefficient was higher than that of the pyro block thickness. On the other hand, the P-values of battery thickness and heating power showed that there was not a statistically significant relationship between these two factors and the response variable.

Table 8

Variance analysis.

Source	DF	Adj SS	Adj MS	F-Value	P-Value
heat transfer coefficient	3	101,056	33685.4	171.65	0.001
battery thickness	3	1901	633.8	3.23	0.181
pyro block thickness	3	5925	1975.1	10.06	0.045
heating power	3	4816	1605.4	8.18	0.059
Error	3	589	196.3		
Total	15	114,288			

5. Conclusion

In the present study, battery preheating for large format prismatic cell operating at lower temperature conditions is numerically investigated with 3 different types of heating methods. Achieved results suggest, heating method 1 is capable to maintain temperature uniformity with a gradient of 3.51°C as compared to other methods, however heating method 1 is not capable to heat up the battery quickly as compared to heating methods 2 and 3. Hence we proposed an improved design for heating method 1 which is capable of heating the battery quickly with the lowest energy consumption by the heater. The following are important conclusions from the present study:

- In sub-zero operating temperatures, it is not only important to heat up the battery quickly but also it is vital to store the generated heat inside the battery volume for longer duration of time. In the present study we are able to achieve this objective by integrating lower thermal conductive pyro liner insulation at the wall of battery. Pyro liner typically have lower thermal conductivity which serves as an optimal insulation capabilities.
- While performing energy analysis of battery preheating system, the heat transfer coefficient (HTC) is an important factor to investigate since shift in the value of HTC can greatly contribute the required amount of energy by heater to preheat the battery. Hence we performed parametric analysis of varying HTC between 5, 10 and 15 W/m²K to verify its contribution over required heating energy and time to preheat optimal battery configuration 1. Monitored results show, for HTC of 5 W/m²K, 624s is required along with 1020 J heating power consumption. Furthermore when HTC is increased, the time required to reach 25°C threshold also increases along with heater energy requirement, as in for HTC of 10 W/m²K, 700s is required with 1150 J and for HTC 15 W/m²K, and 765s is required with 1280 J.
- In electric automotive and energy storage application, the thickness of prismatic battery varies based on storage capacity. Considering the fact, in the present study we had investigated the influence of battery thickness over the required amount of energy by heater along with its transient time duration. The parametric study is done with 39mm, 42mm and 45 mm battery thickness; monitored results show 39mm battery thickness required 624s with 1020 J of energy consumption by the heater. For 42mm battery thickness, the time required is 670s with 1045 J energy consumption by the heater and for 45 mm battery thickness, the time required is 721s with 1065 J of energy consumption.
- In the sub-zero ambient temperature, one of the important objective of the battery preheating is to heat the battery quicker and able to store generated heat inside the battery for longer duration of time. Based on the comparative analysis between generic model and improvised design it is found that the exact time required with the improved model is 400s to reach the temperature threshold limit with 950 J of energy consumption by the heater, comparatively with generic heating method time required to reach threshold limit is 624s with heater energy consumption of 1020 J which is more than the improved model.
- In order to investigate the generated heat storing capacity, the generic model and improved model were compared. It was observed that after the power cut-off, the cell internal temperature at the end of 1000s dropped by 20°C for generic model whereas for improved model it only dropped by 4°C.
- The statistically obtained results demonstrated that the heat transfer coefficient was the most significant factor for the time to reach 25°C. On the other hand, the battery thickness exhibited the lowest impact on the time. The ANOVA table demonstrated that there was a statistically significant relationship between the heat transfer coefficient/pyro block thickness and the time.

Thus, based on the findings of this study, it can be concluded that, in battery preheating thermal design, by incorporating a lower thermally conductive pyro liner at the wall as insulation and the front face uniform heating method, it is possible to provide excellent battery preheating capabilities. This study shows the critical battery preheating insights which can help battery pack manufactures to select optimal heating arrangements in sub-zero ambient temperature regions. In future scope from this study, we recommend the self-heating and temperature rise phenomenon of battery should be considered and simulation capabilities should be extended on pack level configuration with experimental validation support, where objective should be to store generated heat inside battery pack for longer duration when vehicle is parked as a sub-zero battery-soaked condition. We also recommend, for considering self-heating temperature rise phenomenon, standard non-linear vehicle drive cycle from that particular regional sub-zero location should be taken in account for simulation since temperature rise due to battery self-heating is sensitively based on mode of driving and route of driving.

CRedit authorship contribution statement

Virendra Talele: Conceptualization, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **Uğur Moralı:** Software, Visualization, Writing – original draft, Writing – review & editing. **Mahesh Suresh Patil:** Validation, Supervision. **Satyam Panchal:** Validation, Supervision. **Karvinkoppa Mathew:** Validation, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

References

- [1] Talele V, Kore R, Desai H, Chandak A, Sangwan H, Bhale G, Bhirud A, Patrikar S, Nema A, Patil NG. Thermal design and numerical investigation of cold plate for active water cooling for high-energy density lithium-ion battery module. In *Energy Storage Systems 2023* (pp. 343–364). Springer, Singapore.
- [2] V. Talele, P. Zhao, Effect of nano-enhanced phase change material on the thermal management of a 18650 NMC battery pack, *J. Storage Mater.* 1 (64) (2023 Aug), 107068.
- [3] M.K. Tran, A. DaCosta, A. Mevawalla, S. Panchal, M. Fowler, Comparative study of equivalent circuit models performance in four common lithium-ion batteries: LFP, NMC, LMO, NCA, *Batteries* 7 (3) (2021 Sep) 51.
- [4] Talele V, Thorat P, Gokhale YP, Chandak A, Mathew VK. Role of Phase Change Material Thermal Conductivity on Predicting Battery Thermal Effectiveness for Electric Vehicle Application. In *Energy Storage Systems 2023* (pp. 303–342). Springer, Singapore.
- [5] V. Talele, P. Thorat, Y.P. Gokhale, V.K. Mathew, Phase change material based passive battery thermal management system to predict delay effect, *J. Storage Mater.* 15 (44) (2021 Dec), 103482.
- [6] Y. Ji, C.Y. Wang, Heating strategies for Li-ion batteries operated from subzero temperatures, *Electrochim. Acta* 30 (107) (2013 Sep) 664–674.
- [7] R. Jilte, A. Afzal, S. Panchal, A novel battery thermal management system using nano-enhanced phase change materials, *Energy* 15 (219) (2021 Mar), 119564.
- [8] V. Talele, P. Thorat, Y.P. Gokhale, H. Desai, Technical review on battery thermal management system for electric vehicle application, *Energy Storage Syst.* (2023) 177–225.
- [9] A. Mevawalla, S. Panchal, M.K. Tran, M. Fowler, R. Fraser, One dimensional fast computational partial differential model for heat transfer in lithium-ion batteries, *J. Storage Mater.* 1 (37) (2021 May), 102471.
- [10] S. Tippmann, D. Walper, L. Balboa, B. Spier, W.G. Bessler, Low-temperature charging of lithium-ion cells part I: Electrochemical modeling and experimental investigation of degradation behavior, *J. Power Sources* 15 (252) (2014 Apr) 305–316.
- [11] H. Ge, J. Huang, J. Zhang, Z. Li, Temperature-adaptive alternating current preheating of lithium-ion batteries with lithium deposition prevention, *J. Electrochem. Soc.* 163 (2) (2016) A290–A299.
- [12] C.Y. Wang, G. Zhang, S. Ge, T. Xu, Y. Ji, X.G. Yang, Y. Leng, Lithium-ion battery structure that self-heats at low temperatures, *Nature* 529 (7587) (2016 Jan) 515–518.
- [13] J. Liang, Y. Gan, M. Tan, Y. Li, Multilayer electrochemical-thermal coupled modeling of unbalanced discharging in a serially connected lithium-ion battery module, *Energy* 15 (209) (2020 Oct), 118429.
- [14] S.D. Chitta, C. Akkaldevi, J. Jaidi, S. Panchal, M. Fowler, R. Fraser, Comparison of lumped and 1D electrochemical models for prismatic 20Ah LiFePO4 battery sandwiched between minichannel cold-plates, *Appl. Therm. Eng.* 25 (199) (2021 Nov), 117586.
- [15] J. Duan, J. Zhao, X. Li, S. Panchal, J. Yuan, R. Fraser, M. Fowler, Modeling and analysis of heat dissipation for liquid cooling lithium-ion batteries, *Energies* 14 (14) (2021 Jan) 4187.
- [16] S. Al Hallaj, J.R. Selman, A novel thermal management system for electric vehicle batteries using phase-change material, *J. Electrochem. Soc.* 147 (9) (2000 Sep 1) 3231.
- [17] A.P. Sasmito, T. Shamim, A.S. Mujumdar, Passive thermal management for PEM fuel cell stack under cold weather condition using phase change materials (PCM), *Appl. Therm. Eng.* 58 (1–2) (2013 Sep 1) 615–625.
- [18] Z. Ling, X. Wen, Z. Zhang, X. Fang, T. Xu, Warming-up effects of phase change materials on lithium-ion batteries operated at low temperatures, *Energ. Technol.* 4 (9) (2016 Sep) 1071–1076.
- [19] G. Zhong, G. Zhang, X. Yang, X. Li, Z. Wang, C. Yang, C. Yang, G. Gao, Researches of composite phase change material cooling/resistance wire preheating coupling system of a designed 18650-type battery module, *Appl. Therm. Eng.* 25 (127) (2017 Dec) 176–183.
- [20] Z. Lei, C. Zhang, J. Li, G. Fan, Z. Lin, Preheating method of lithium-ion batteries in an electric vehicle, *J. Mod Power Syst. Clean Energy* 3 (2) (2015 Jun) 289–296.
- [21] L. Li, C. Xu, R. Chang, C. Yang, C. Jia, L. Wang, J. Song, Z. Li, F. Zhang, B. Fang, X. Wei, H. Wang, Q. Wu, Z. Chen, X. He, X. Feng, H. Wu, M. Ouyang, Thermal-responsive, super-strong, ultrathin firewalls for quenching thermal runaway in high-energy battery modules, *Energy Storage Mater.* 40 (2021) 329–336.
- [22] Y. Gan, J. Wang, J. Liang, Z. Huang, M. Hu, Development of thermal equivalent circuit model of heat pipe-based thermal management system for a battery module with cylindrical cells, *Appl. Therm. Eng.* 5 (164) (2020 Jan), 114523.
- [23] J. Liang, Y. Gan, Y. Li, M. Tan, J. Wang, Thermal and electrochemical performance of a serially connected battery module using a heat pipe-based thermal management system under different coolant temperatures, *Energy* 15 (189) (2019 Dec), 116233.
- [24] X. Dai, D. Kong, J. Du, Y. Zhang, P. Ping, Investigation on effect of phase change material on the thermal runaway of lithium-ion battery and exploration of flame retardancy improvement, *Process Saf. Environ. Prot.* 1 (159) (2022 Mar) 232–242.
- [25] W. Zhang, Z. Liang, X. Yin, G. Ling, Avoiding thermal runaway propagation of lithium-ion battery modules by using hybrid phase change material and liquid cooling, *Appl. Therm. Eng.* 5 (184) (2021 Feb), 116380.
- [26] J. Liang, Y. Gan, Y. Li, Investigation on the thermal performance of a battery thermal management system using heat pipe under different ambient temperatures, *Energ. Convers. Manage.* 1 (155) (2018 Jan) 1–9.
- [27] J. Wang, Y. Gan, J. Liang, M. Tan, Y. Li, Sensitivity analysis of factors influencing a heat pipe-based thermal management system for a battery module with cylindrical cells, *Appl. Therm. Eng.* 25 (151) (2019 Mar) 475–485.
- [28] Y. Wang, Z. Rao, S. Liu, X. Li, H. Li, R. Xiong, Evaluating the performance of liquid immersing preheating system for Lithium-ion battery pack, *Appl. Therm. Eng.* 25 (190) (2021 May), 116811.
- [29] Z.G. Qu, Z.Y. Jiang, Q. Wang, Experimental study on pulse self-heating of lithium-ion battery at low temperature, *Int. J. Heat Mass Transf.* 1 (135) (2019 Jun) 696–705.
- [30] V. Talele, M.S. Patil, S. Panchal, R. Fraser, M. Fowler, S.R. Gunti, Novel metallic separator coupled composite phase change material passive thermal design for large format prismatic battery pack, *J. Storage Mater.* 1 (58) (2023 Feb), 106336.
- [31] Deshmukh A, Talele V, Chandak A. The Role of Lower Thermal Conductive Refractory Material in Energy Management Application of Heat Treatment Furnace. In *Energy Storage Systems 2023* (pp. 21–37). Springer, Singapore.
- [32] Website link (03/07/2023) – <https://himaxelectronics.com/product-item/lifepo4-battery-cell-3-2v-150ah/>.
- [33] W. Yang, F. Zhou, Y. Liu, X. Chen, Preheating performance by heating film for the safe application of cylindrical lithium-ion battery at low temperature, *Fire Technol.* 23 (2022 Apr) 1–21.
- [34] Website link (03/07/2023) - https://www.morganthermalceramics.com/media/3257/16518-pyro-bloc-module-manual_06212016.pdf.
- [35] A. Kurhade, V. Talele, T.V. Rao, A. Chandak, V.K. Mathew, Computational study of PCM cooling for electronic circuit of smart-phone, *Mater. Today: Proc.* 1 (47) (2021 Jan) 3171–3176.
- [36] A.R. Sharma, C.S. Sai, A. Kumar, R.V. Reddy, D. Danyharsha, R. Jilte, Three-dimensional CFD study on heat dissipation in cylindrical lithium-ion battery module, *Mater. Today: Proc.* 1 (46) (2021 Jan) 10964–10968.
- [37] K. Zou, X. Chen, Z. Ding, J. Gu, S. Lu, Jet behavior of prismatic lithium-ion batteries during thermal runaway, *Appl. Therm. Eng.* 1 (179) (2020 Oct), 115745.
- [38] A.S. Mussa, G. Lindbergh, M. Klett, P. Gudmundson, P. Svens, R.W. Lindström, Inhomogeneous active layer contact loss in a cycled prismatic lithium-ion cell caused by the jelly-roll curvature, *J. Storage Mater.* 1 (20) (2018 Dec) 213–217.
- [39] V. Talele, M.S. Patil, S. Panchal, R. Fraser, M. Fowler, Battery thermal runaway propagation time delay strategy using phase change material integrated with pyro block lining: Dual functionality battery thermal design, *J. Storage Mater.* 15 (65) (2023 Aug), 107253.

- [40] J. Zhang, H. Liu, M. Zheng, M. Chen, L. Zhao, D. Du, Numerical study on a preheating method for lithium-ion batteries under cold weather conditions using phase change materials coupled with heat films, *J. Storage Mater.* 1 (47) (2022 Mar), 103651.
- [41] J. Liang, Y. Gan, M. Yao, Y. Li, Numerical analysis of capacity fading for a LiFePO₄ battery under different current rates and ambient temperatures, *Int. J. Heat Mass Transf.* 1 (165) (2021 Feb), 120615.
- [42] Electric bus safety requirement for heating and insulation Amendment 38032. (01/01/2022) Weblink – <https://www.codeofchina.com/standard/GB38032-2020.html>.
- [43] A.S. Kurhade, G. Murali, Thermal control of IC chips using phase change material: A CFD investigation, *Int. J. Mod. Phys. C* 33 (12) (2022 Dec 28) 2250159.
- [44] Website link (03/07/2023) - [rcf-pyroblocmodules_eng.pdf](#) (morganthermalceramics.com).
- [45] U. Morali, Computational modeling and statistical evaluation of thermal behavior of cylindrical lithium-ion battery, *J. Storage Mater.* 55 (2022 Nov), 105376.
- [46] U. Morali, A numerical and statistical implementation of a thermal model for a lithium-ion battery, *Energy* 240 (2022 Feb), 122486.